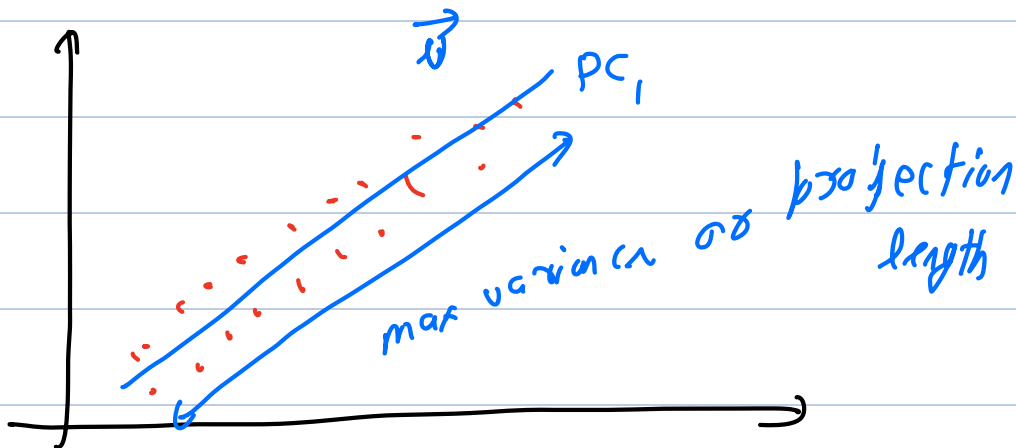
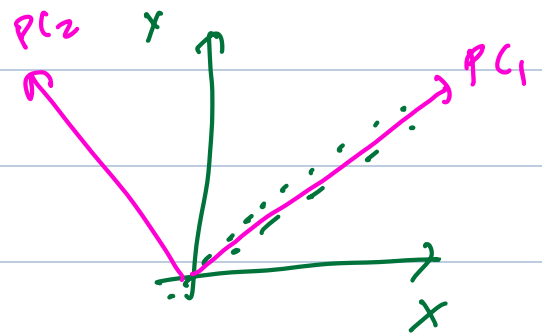


PCA-2 & T-SNE



$$V \cdot \vec{v} = \lambda \cdot \vec{v}$$

eigen vector
eigen value



3 columns
of
3D data $\xrightarrow{\text{PCA}}$ $\vec{v}_1, \vec{v}_2, \vec{v}_3$
 $\lambda_1, \lambda_2, \lambda_3$

$$V = \frac{X^T X}{N}$$

$$V = \text{np.matmul}(x^T, x) / N$$

$$V = np \cdot \text{cov}(x)$$

$$v, \text{lambda} = np \cdot \text{linalg} \cdot \text{eig}(V)$$

5 dim
(5 features)

PCA →

5 PC

→ 10

→ 7

→ 6

→ 3

→ 1

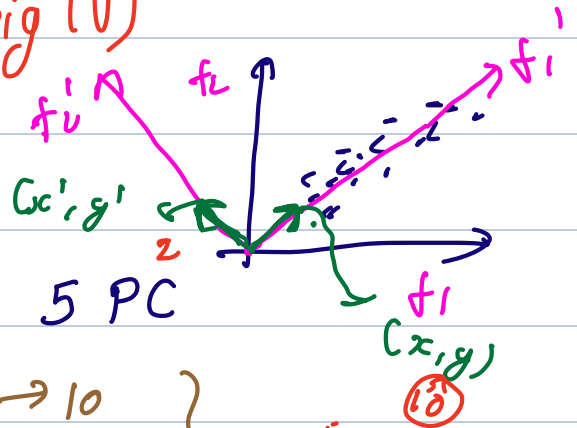
Eigen
value

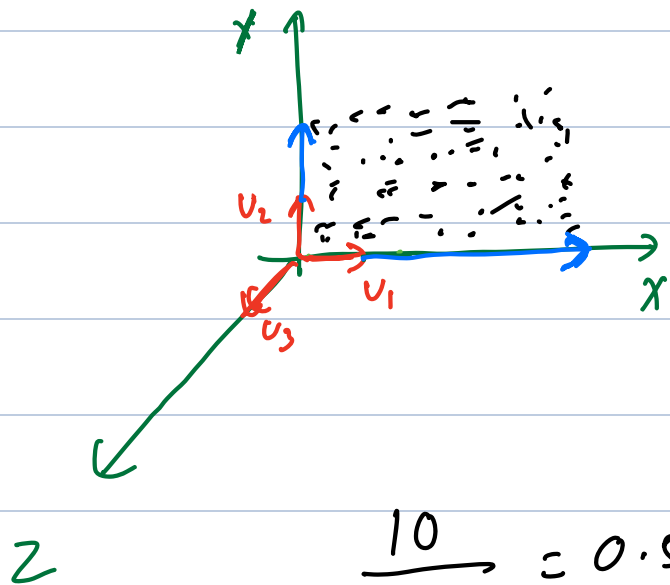
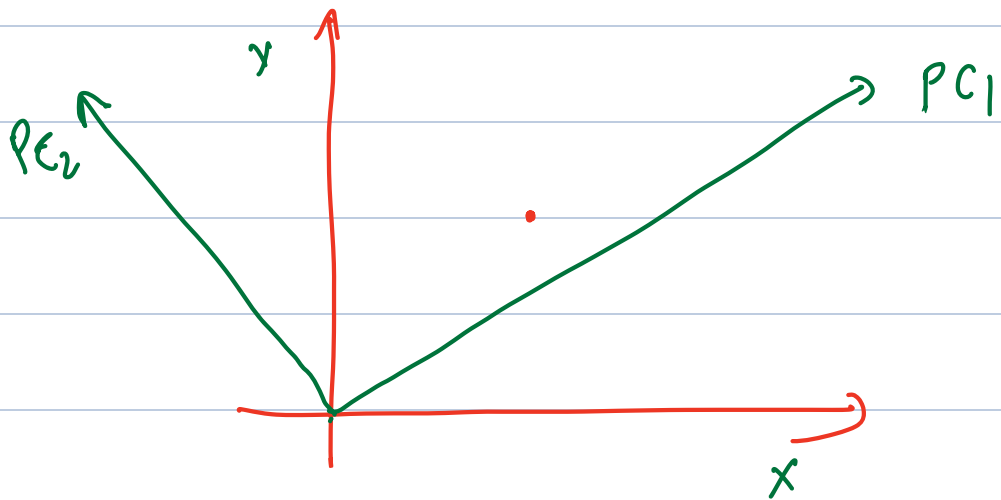
$$\frac{10}{10 + 7 + 6 + 3 + 1}$$

$$\frac{10}{27} = 0.37$$

$$\frac{10 + 7 + 6}{27} = \frac{23}{27} \approx 84\%$$

$$\frac{10 + 7}{27} = \frac{17}{27} = 63$$





$$\frac{10}{17} = 0.58$$

$$\vec{u}_1 \rightarrow \lambda_1 = 10$$

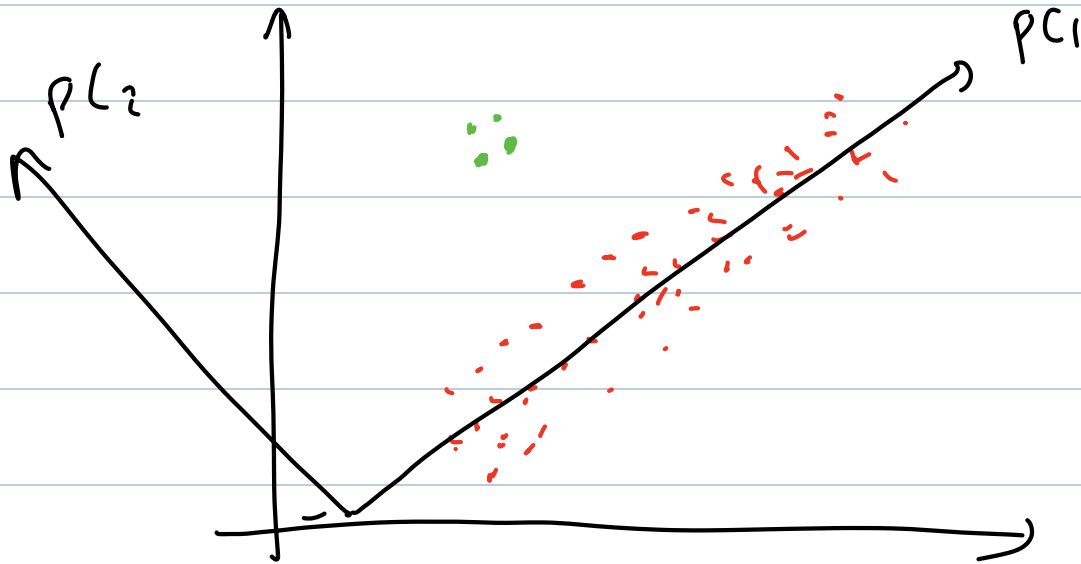
$$\vec{u}_2 \rightarrow \lambda_2 = 6$$

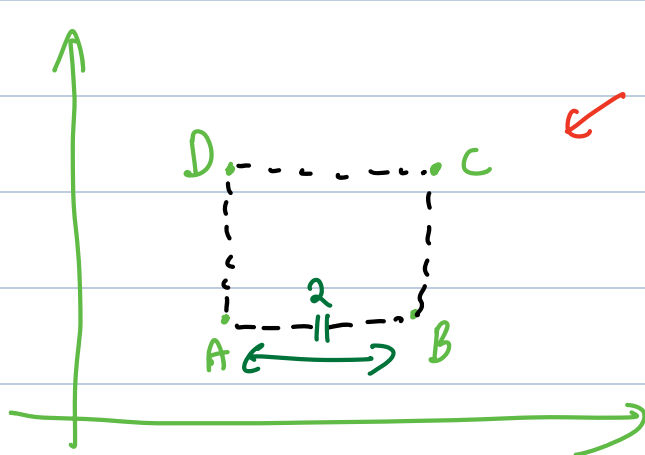
$$\vec{u}_3 \Rightarrow \lambda_3 = 1$$

$$\frac{10 + 6}{17} = \frac{16}{17} = 0.94$$

T-SNE

* - distributed stochastic Neighbourhood Embedding

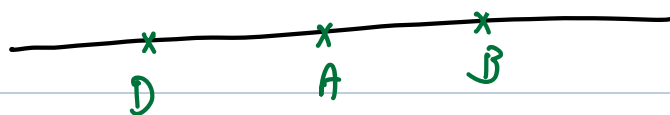




← Crowding Problem

$$P_N(A, B) = 1$$

$$Q_N(A, B) = 1$$



$$P(A, B) = 1$$

↓

$$d(A, B) = 1$$

$$P(A, B) = \frac{1}{d(A, B)}$$

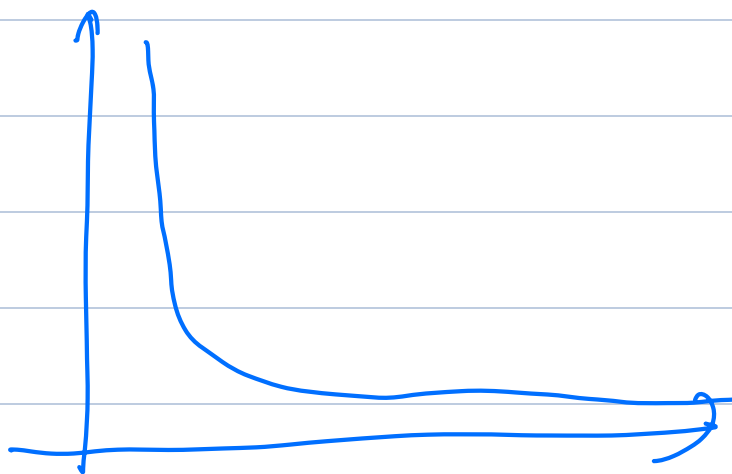
x x

→ euc

$$= \frac{1}{\|A - B\|_2} \rightarrow L_2 - \text{Norm}$$

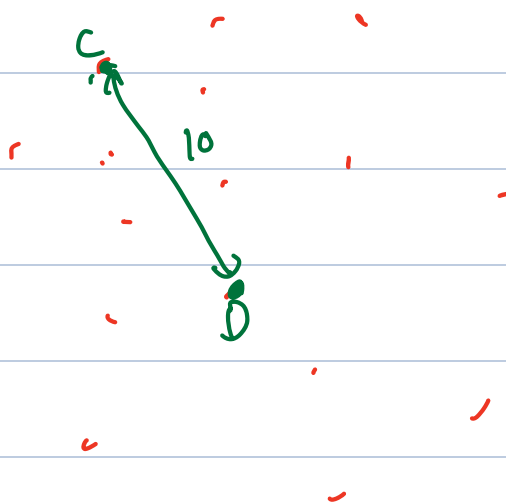
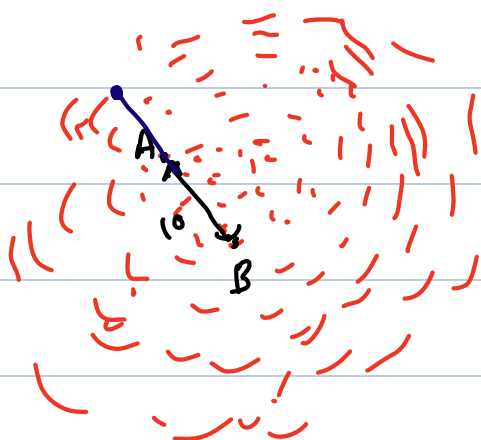
$$A \& B \rightarrow 0$$

$$P \rightarrow \infty$$



$$A - B = 2 \quad \Rightarrow \quad P = \frac{1}{2}$$

$$A - B' = 4 \quad \Rightarrow \quad P = \frac{1}{4}$$



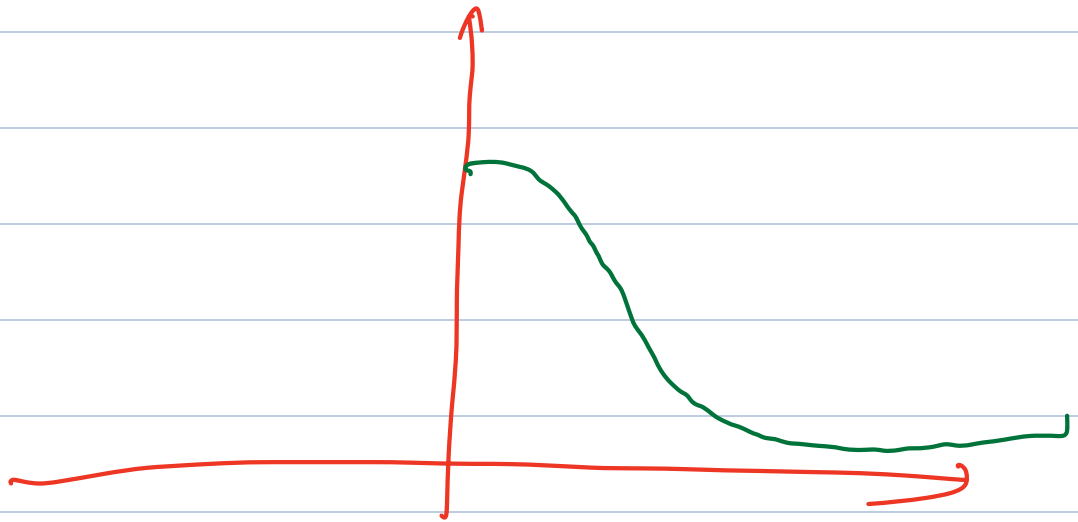
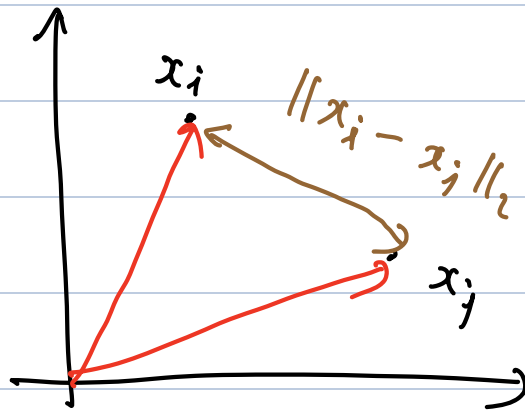
set 1
 $\downarrow$
 MCI $\rightarrow$ 8.0, 8.1, 8.3, 8.2, 8.9, 8.6, ..., 9.0

set 2
 $\downarrow$
 MCI $\rightarrow$ 8.0, 8.5, 9.0, —————



$$P_N(x_i, x_j) \propto \frac{1}{d_{ij}}$$

$$\propto \frac{1}{\|x_i - x_j\|_2}$$



$P = \frac{1}{d}$ $\times$ $\rightarrow$ Graph $\times$
 $\downarrow$
 created graph
 $\Downarrow$
 for such

- Sumit Shukla
- Ekta shah
- SQL revamp
- Assignment new style : Nikhil & MC revamp
- Slow batch
 - Browtux
 - 2.0 course Pause
 - Module test
 -

Suraaj ownership



1100 →

← ①

Clearance Rate

②

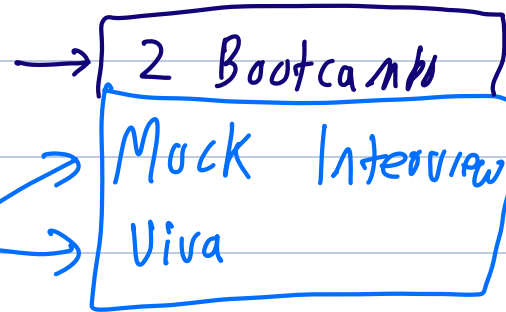
NPS → Mohit

③

Content (Domain Analysis)

DS overview

GenAI)



④ Resume (Guidelines)

121

→ Aditya

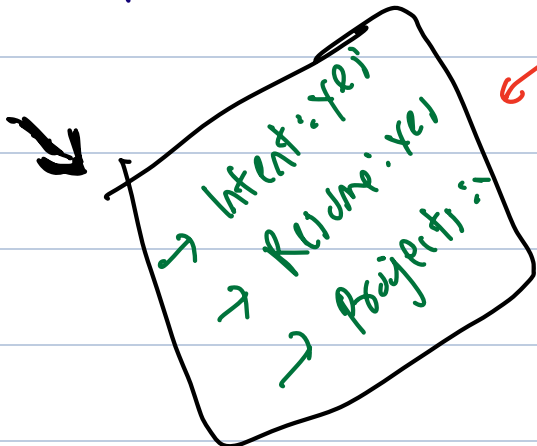
→ RI: SQL + Python & EDA

1100 : 700

communication problem

Application is an issue

Net offers



→ Cumulative Test

Cumulative Test

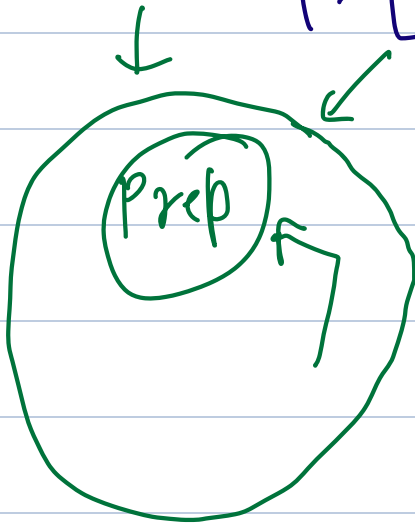
→ Bootcamps

D6

→ DA

→ 700
↓
→ 115 ✓

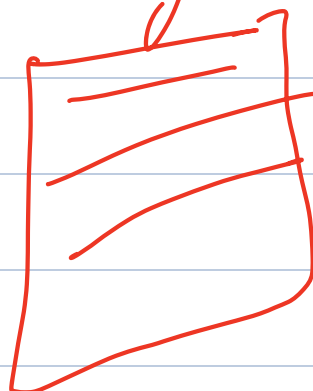
→ 600 ✓
→ Why failed?
→ Bootcamps



- ① Pause for learners
- ②

Aman

Talking to



Shout!

Track

→ FP → sample talk to learners

one Topic issue

→ Relevant Job

→

Analysis ←

7 years

CTC: 20.70

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